
Need for container orchestration tool
Trend from Monolith to Microservices
Increased usage of containers
Container
App
Container
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Container
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Container
App

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Need for container orchestration tool
Trend from Monolith to Microservices
Increased usage of containers
Demand for a proper way of managing those hundreds of containers

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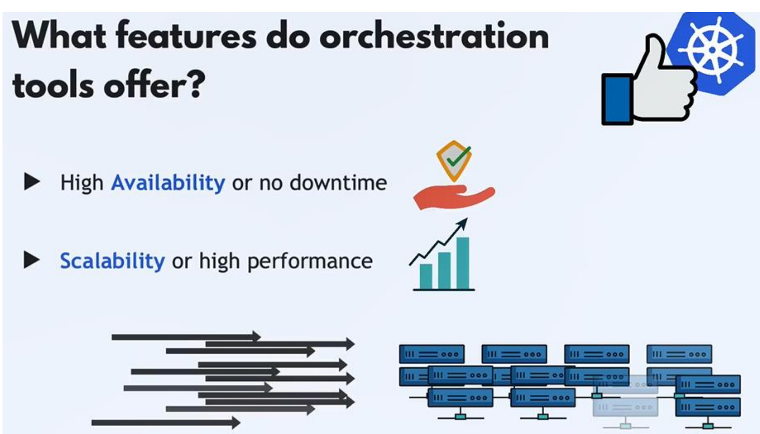
What features do orchestration tools offer?
High Availability or no downtime
Scalability or high performance

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Kubernetes contains one master node, and other nodes (workers) that can connect to master

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Note: *Kubelet* is program that make the nodes in the cluster reach to each other , and it's responsible of running the applications *inside the nodes* (from me: we can imagine it as *replica of app* with API Gateway)

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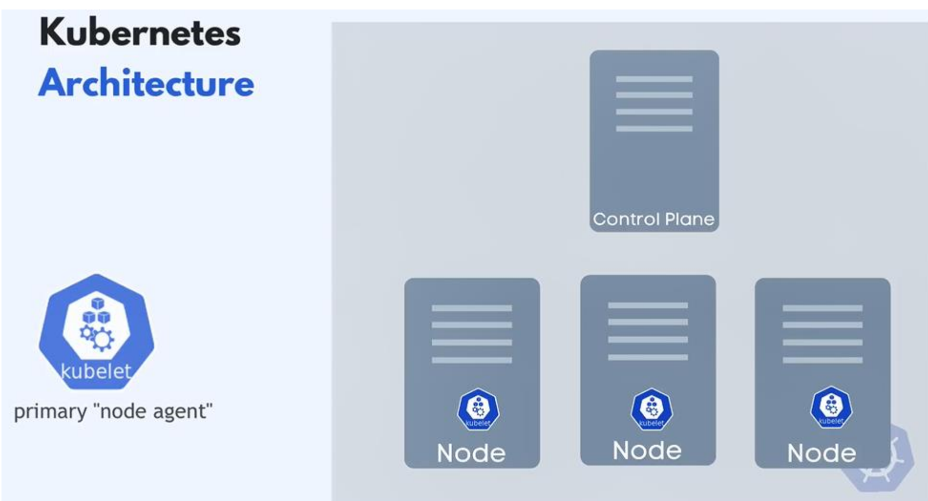

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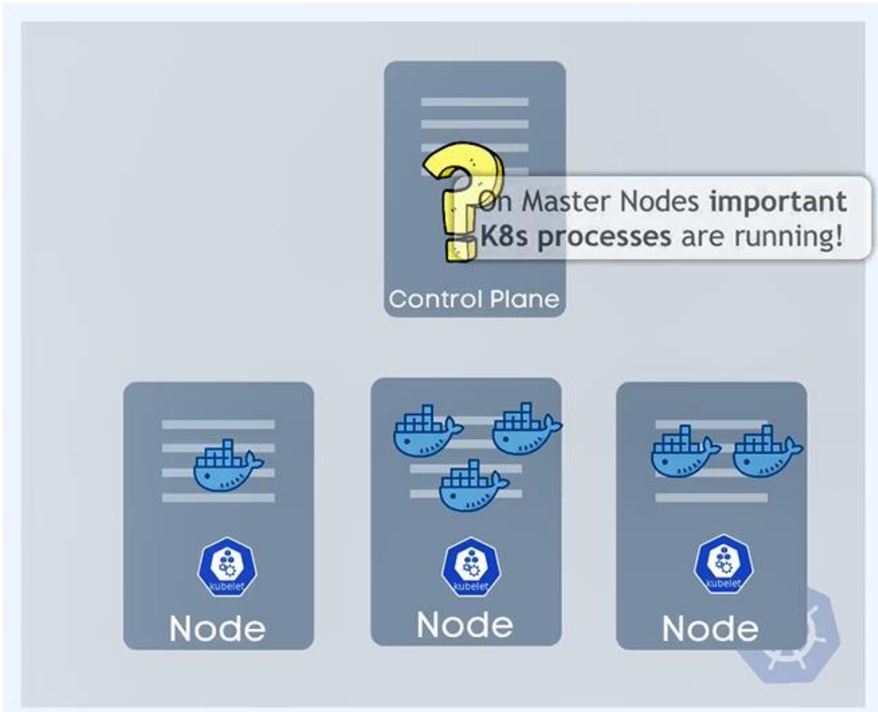
On Master Nodes important K8s processes are running!
Control Plane
Node
Node
Node

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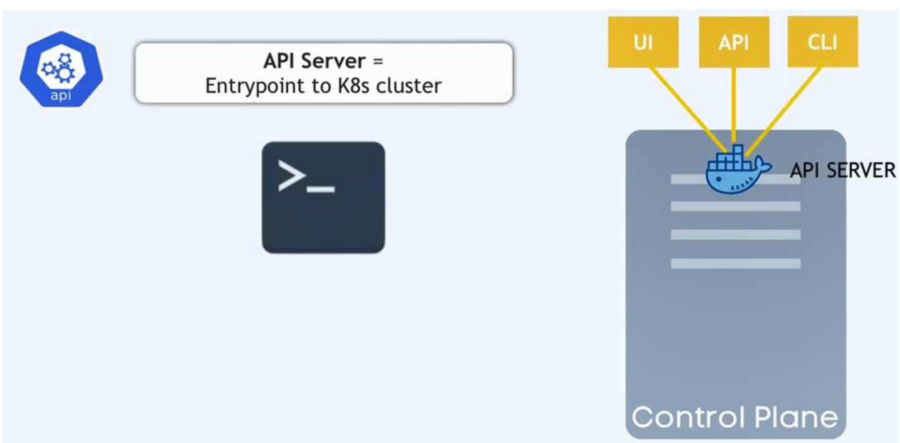
API Server =
Entrypoint to K8s cluster
UI
API
CLI
API SERVER
Control Plane

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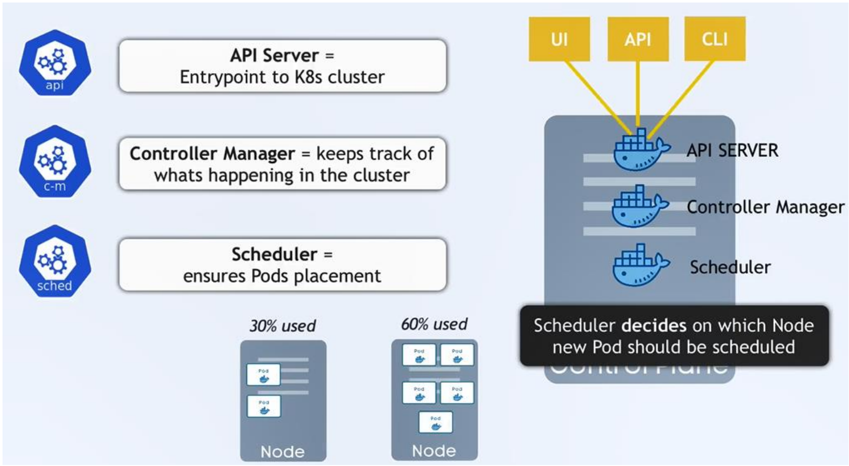

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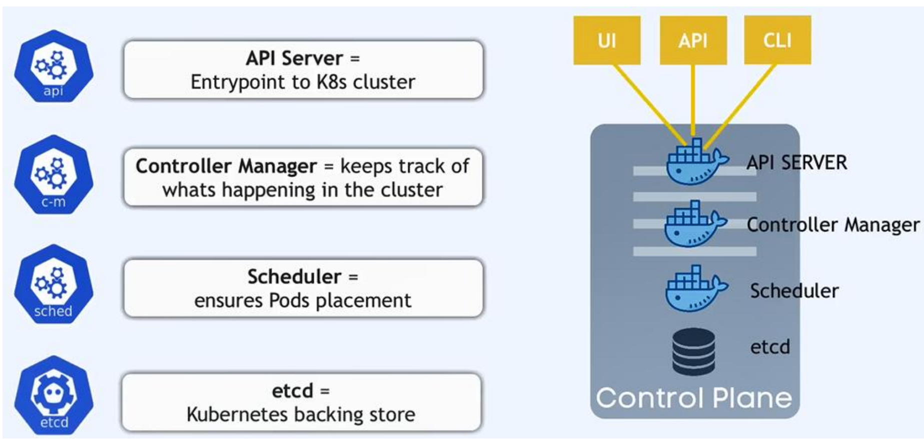

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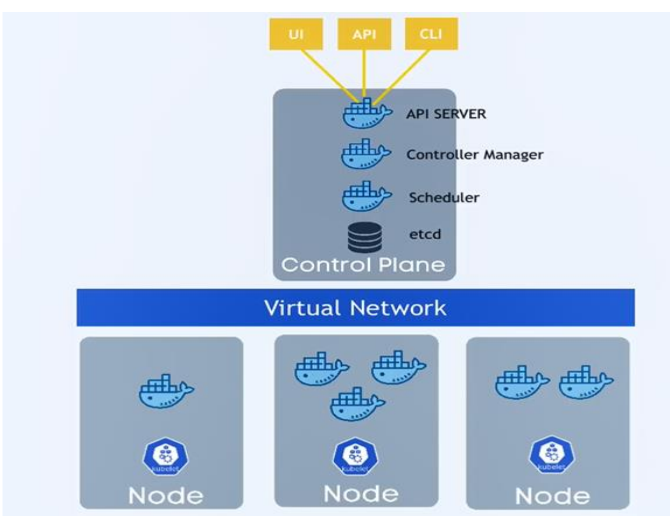
Virtual Network: allow nodes to talk to each other



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Node
Node
Node
Worker Nodes
higher workload
much bigger and more resources

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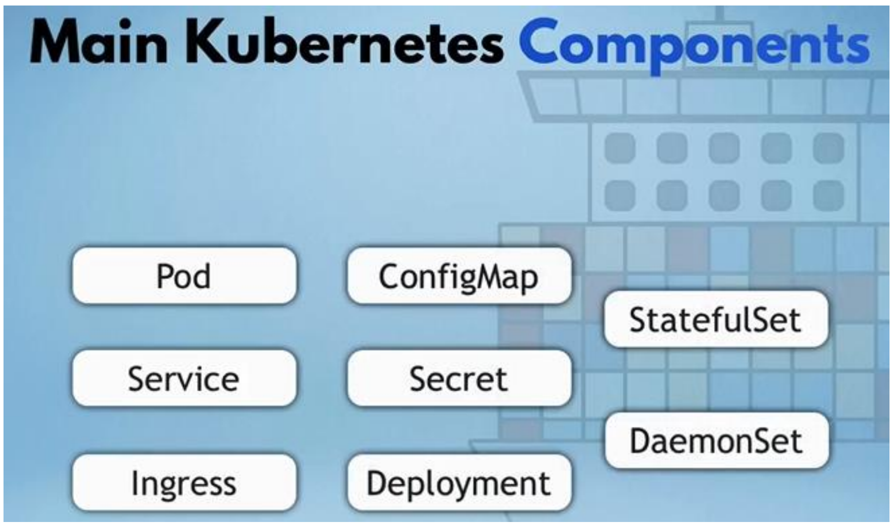
Main Kubernetes Components
Pod
ConfigMap
StatefulSet
Service
Secret
DaemonSet
Ingress
Deployment

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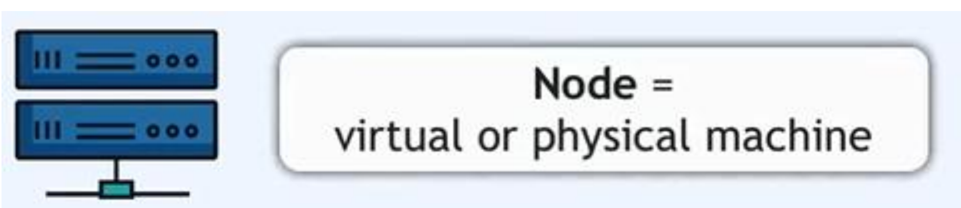
Node =
virtual or physical machine

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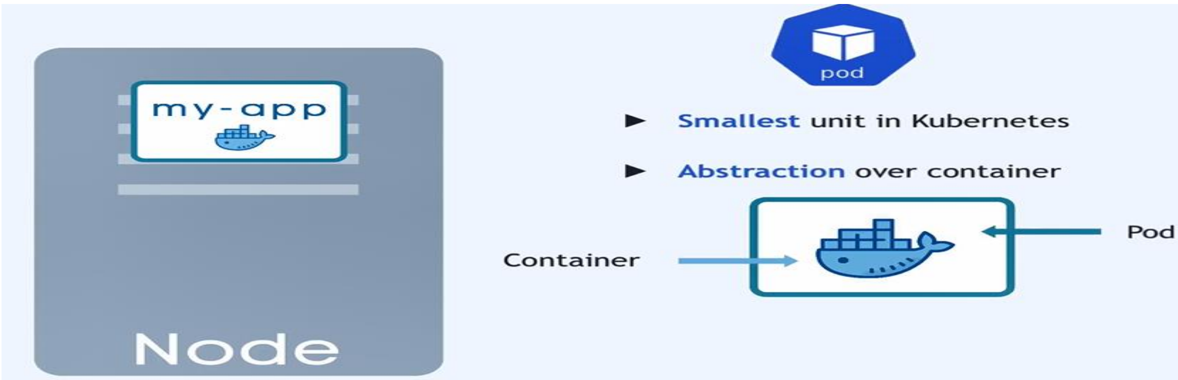

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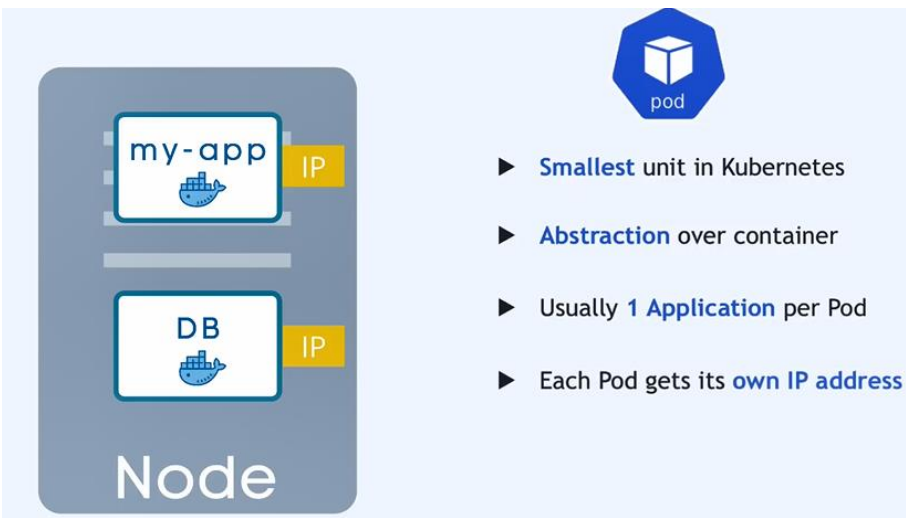
Not the container, the pod get its own ip-addr:



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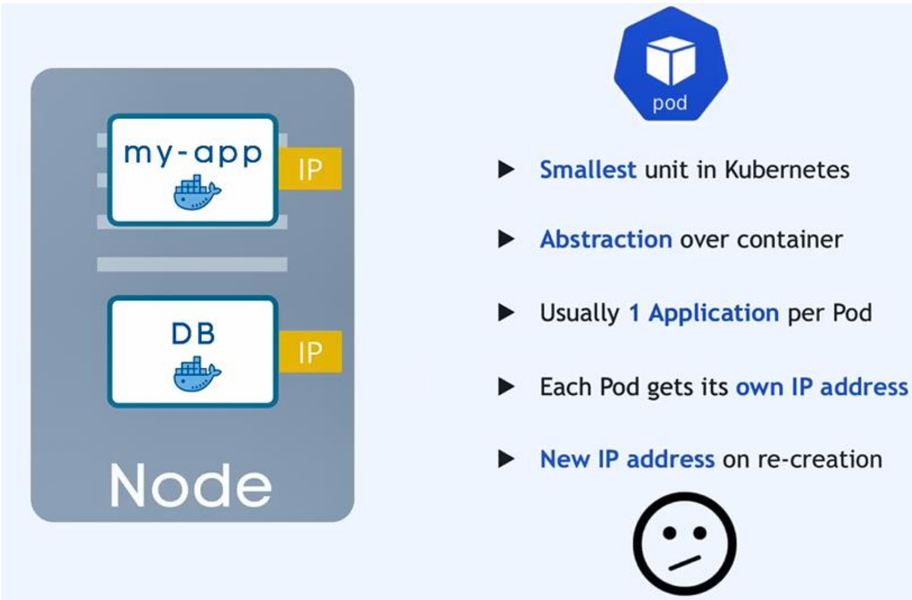
If any pod is dying for any reason then new ip-addr will assigned:



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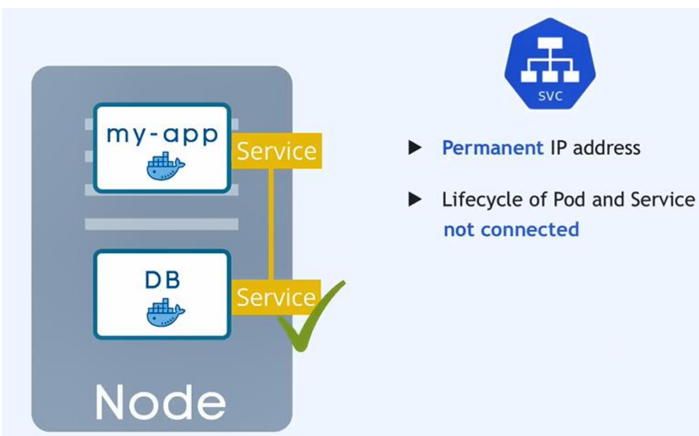
To solve the previous problem we use services (even if the pod die the service will still have the same ip-addr):



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External service: the service that open the communication with external resources (ex: from node/cluster to our browser)

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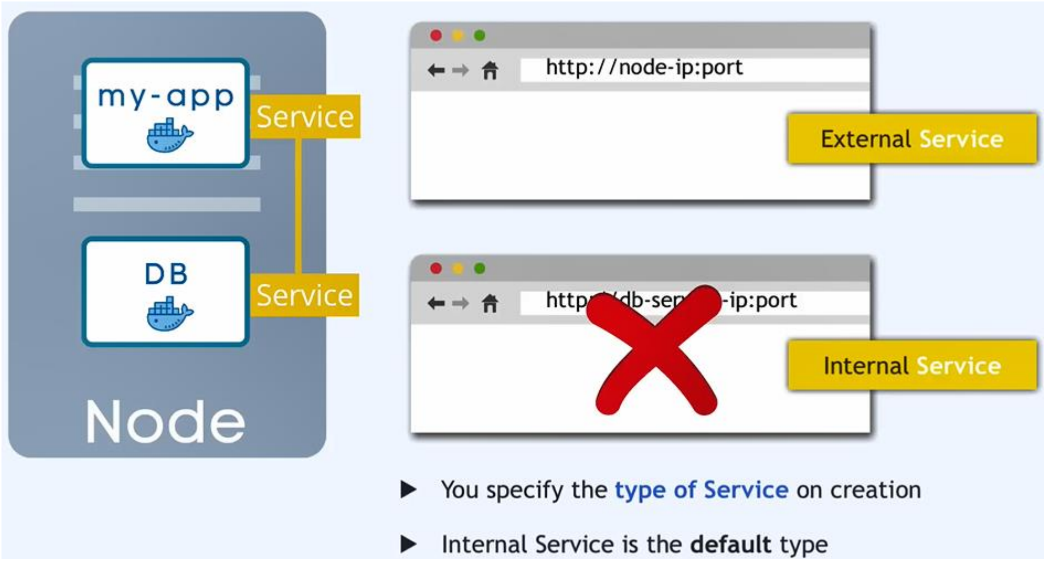

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Ingress: like service but it routes the request then call the right service

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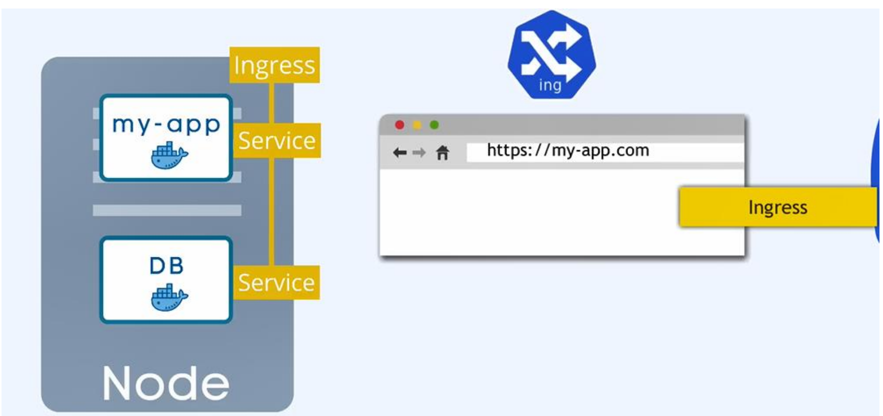

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The ***main advantages of services*** is to make the pods communicate with each other

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The ***advantages of ConfigMap*** in ***kubernetes*** is when we change the configurations of our app, we don't need to rebuild our image from start

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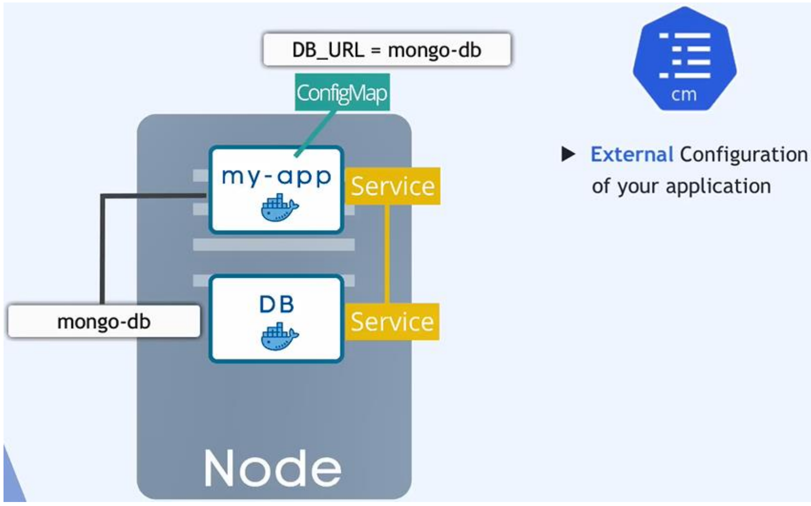

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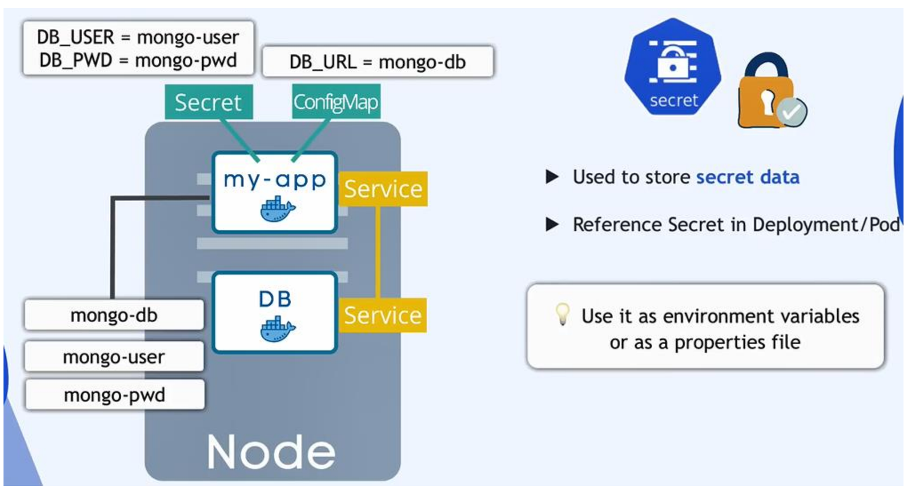
To store secret data we can use Secret, but it store them as base64, not in encrypted way, we must enable the encryption, and we must enable RBAC



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The another problem of pod is, when the pod restart it will loss the data

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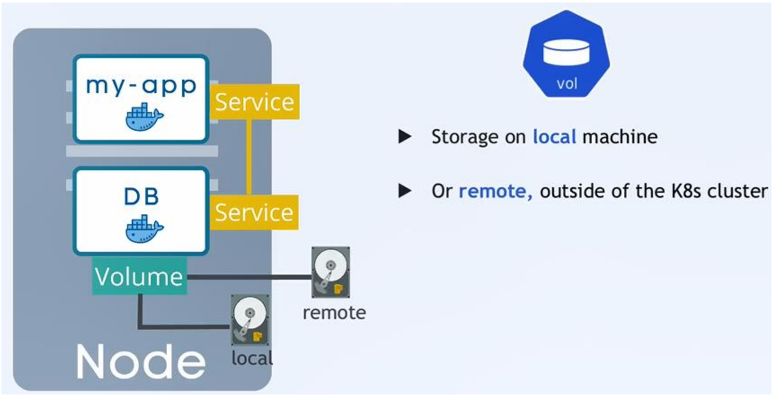
my-app
Service
DB
Service
Volume
remote
local
Node
vol
Storage on local machine
Or remote, outside of the K8s cluster

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